

CANONICAL LESSON

Maintain First, Replace With Evidence

Decision-centred foundations for accountable sector intelligence

Production Unit	PU-B07-C56
Notebook	V7_D_N06
Status	Controlled educational draft v0.1
Data boundary	Synthetic instructional data only

Central decision: Prioritize infrastructure maintenance and replacement using condition, service criticality, failure probability, consequence, lifecycle cost, uncertainty, equity, and accountable asset authority.

Learning outcomes

Why this matters

Replacing the most visible asset first can waste resources while less visible critical failures remain unaddressed.

The analytical workflow is useful only when evidence, uncertainty, authority, and action are connected explicitly. The model supports professional judgement; it does not replace accountable institutional authority.

Worked cross-sector example

A municipality compares preventive maintenance, rehabilitation, and replacement scenarios for a deteriorating water asset.

Boundary: The example supports learning and review. It does not constitute an official statistic, eligibility determination, enforcement action, clinical judgement, or operational authorization.

Practice and reflection

- 1.
- Add network dependencies.
- 2.
- Compare preventive and reactive costs.

- 3.
- Explain the safety boundary.

Answer key

- 1.
- Model upstream/downstream consequences and single points of failure.
- 2.
- Use lifecycle cost, failure probability, service disruption, and uncertainty.
- 3.
- Only qualified inspectors and responsible authorities can certify, restrict use, procure, or authorize works.

Operational readiness checklist

- Purpose and decision owner are named.
- Evidence definitions, provenance, and lawful access are documented.
- Uncertainty and abstention are visible.
- Human review, escalation, monitoring, incident response, and retirement are assigned.
- Equity, accessibility, security, and sector-specific approval are complete.